

# Cyber Security Bootcamp

Rework Academy - Versi Detail

| Program            | Pembelajaran | Output                      |
|--------------------|--------------|-----------------------------|
| Online - 12 minggu | 20 sesi live | Assessment project & report |

Setiap section berikut memuat ringkasan pembelajaran dan rincian topik yang dicantumkan pada halaman program. Istilah teknis dipertahankan untuk menjaga kesesuaian dengan kurikulum resmi.

## Fondasi & Etika

### 01 Cybersecurity Fundamentals & Ethical Security Practice

*Peserta mempelajari dasar cybersecurity, threat landscape, attacker mindset, attack surface, istilah keamanan, serta batasan legal dalam praktik security testing profesional.*

- Ekosistem cybersecurity dan modern threat landscape
- Perlindungan website, API, sistem autentikasi, dan infrastruktur
- Perbedaan vulnerability, exploit, threat, risk, dan business impact
- Tipe threat actor dan pola pikir attacker
- Attack surface: login, endpoint, parameter, serta upload
- Istilah inti seperti payload, session, logs, dan authorization
- Ethical hacking, authorization, scope, serta responsible disclosure

### 02 Red Team, Blue Team & Purple Team Workflow

*Peserta memahami peran Red Team, Blue Team, dan Purple Team pada operasi keamanan: dari pengujian ofensif hingga monitoring, remediation, dan kolaborasi.*

- Workflow Red Team untuk offensive security
- Reconnaissance, validasi kerentanan, eksploitasi, dan pelaporan
- Workflow Blue Team untuk monitoring dan detection
- Analisis aktivitas mencurigakan melalui logs dan system visibility
- Purple Team sebagai penghubung praktik ofensif dan defensif
- Penyusunan alur attack, detection, dan remediation
- Penetration testing lifecycle serta rules of engagement

### 03 Technical Foundations for Modern Security Testing: Network, Web, HTTP & API

*Peserta mempelajari fondasi aplikasi web, HTTP, API, jaringan, DNS, dan infrastruktur untuk memahami cara kerja sistem modern dan attack surface-nya.*

- Struktur frontend, backend, database, dan API
- Dashboard, admin panel, authentication system, dan API service
- HTTP request-response, method, header, cookie, dan status code
- Analisis request dan response dengan browser tools
- Endpoint, parameter, token, serta JSON dalam komunikasi API
- Domain, DNS, IP address, port, protocol, dan service
- DNS lookup serta traffic inspection dasar

### 04 Linux Fundamentals for Cybersecurity Operations

*Peserta mempelajari Linux sebagai lingkungan penting untuk cybersecurity, meliputi filesystem, command line, permission, process, logs, dan workflow security testing.*

- Linux environment untuk cybersecurity workflow
- Filesystem, directory structure, dan file management
- Navigasi terminal dan command-line dasar
- File inspection dan process management
- Permission, user privilege, process, serta logs
- Log inspection untuk monitoring dan troubleshooting
- Kaitan Linux dengan reconnaissance dan security tools

# Pengujian Aplikasi & Infrastruktur

## 05 Web Testing Workflow: HTTP, Authentication & Burp Suite

*Peserta mempelajari alur autentikasi, session, cookie, analisis request, Burp Suite interception, serta dasar OWASP Top 10 dalam web security testing.*

- Login workflow, session, cookie, dan authenticated request
- Cara browser mempertahankan authenticated session
- Request inspection dengan Burp Suite dan Chrome DevTools
- Parameter, header, body, cookie, serta response structure
- Request interception dan authentication persistence
- Perbedaan hashing, encoding, dan encryption
- OWASP Top 10 dan vulnerability fundamentals

## 06 Web Vulnerability Discovery & Attack Surface Analysis

*Peserta mempelajari cara mengidentifikasi attack surface, melakukan vulnerability mapping, menilai fitur yang terbuka, dan membangun exploitation mindset pada aplikasi web.*

- Attack surface discovery pada aplikasi web
- Login form, endpoint, parameter, upload, dan exposed feature
- Vulnerability mapping berdasarkan behaviour dan functionality
- Error message, response behaviour, dan application flow
- Entry point pada upload, search, API, dan authentication
- Weak validation dan insecure functionality exposure
- Attack path berdasarkan reconnaissance findings

## 07 Reconnaissance & Information Gathering Workflow

*Peserta mempelajari reconnaissance untuk mengumpulkan informasi target yang berizin, menemukan service terbuka, memetakan infrastruktur, dan menganalisis public exposure.*

- Passive dan active reconnaissance dalam penetration testing
- Target profiling dan public information gathering
- Keterkaitan reconnaissance dengan attack surface mapping
- Subdomain enumeration menggunakan reconnaissance tools
- Service discovery dan analisis exposed endpoint
- Port scanning dan network reconnaissance dengan Nmap
- GitHub exposure, credential leakage, dan pola data sensitif

## 08 API Security, Token & Sensitive Data Exposure Analysis

*Peserta mempelajari API workflow, endpoint, token, session, sensitive data exposure, request manipulation, serta risiko salah konfigurasi pada aplikasi modern.*

- API workflow dan endpoint communication
- Endpoint, parameter, token, dan JSON request-response
- Endpoint analysis serta request inspection dasar
- Token, session, dan authentication data workflow
- Sensitive data exposure dan insecure token handling
- Request analysis serta parameter manipulation
- API exposure, misconfiguration, dan risiko unauthorized access

## 09 SQL Injection Testing & Risk Analysis

*Peserta mempelajari SQL Injection, manipulasi query, bypass autentikasi, pengujian parameter rentan, dan dampak akses database tanpa otorisasi.*

- SQL Injection fundamentals dan database workflow
- Insecure input validation serta query processing
- Testing di vulnerable web environment yang terkontrol
- Query manipulation dan application behaviour
- Authentication bypass pada login workflow
- Payload testing pada parameter rentan
- Dampak SQL Injection terhadap data dan business risk

## 10 Cross-Site Scripting (XSS) Attacks

*Peserta mempelajari XSS, browser rendering, reflected dan stored XSS, kelemahan validasi input, perilaku payload, serta risiko eksekusi script pada browser korban.*

- XSS fundamentals dan browser rendering workflow
- HTML processing dan JavaScript execution
- Payload XSS sederhana pada lingkungan latihan
- Reflected XSS dan kelemahan input validation
- Payload testing pada reflected XSS environment
- Stored XSS dan persistent payload behaviour
- Browser behaviour serta security risk dari XSS

## 11 Authentication, Authorization & Access Control Testing

*Peserta mempelajari pengujian authentication, authorization, access control, IDOR, session manipulation, dan access validation untuk mengenali risiko akses tanpa izin.*

- Authentication workflow dan session validation
- Login request analysis serta session inspection
- Authorization, role, user privilege, dan access control
- Weak authorization dan risiko privilege misuse
- IDOR dan object reference validation
- Parameter manipulation untuk unauthorized access testing
- Session manipulation serta access validation behaviour

## 12 Advanced SQL Injection & Automation-Assisted Testing

*Peserta mempelajari blind dan time-based SQL Injection, SQLMap, automation-assisted testing, validasi manual, false positive, serta dampak eksploitasi tingkat lanjut.*

- Blind SQL Injection dan response-based analysis
- Time-based SQL Injection serta timing observation
- Request inspection, payload testing, dan response validation
- SQLMap untuk automation-assisted testing
- Kaitan manual validation dengan automation tools
- False positive dan verification workflow
- Advanced SQL Injection impact dan business risk

### 13 File Upload Exploitation & Remote Code Execution (RCE) Advance

*Peserta mempelajari file upload workflow, kelemahan validasi, dangerous upload, upload-to-RCE, insecure file handling, dan dampaknya pada server.*

- Alur upload hingga backend processing
- File extension, MIME type, file size, dan content validation
- Weak validation dan insecure file handling
- Dangerous file upload serta payload behaviour
- Hubungan upload weakness dengan dangerous execution
- Upload-to-RCE workflow fundamentals
- Server compromise dan unauthorized access risk

### 14 SSRF, Cloud Exposure & Modern Attack Surface

*Peserta mempelajari SSRF, backend request workflow, cloud exposure, metadata service, internal service exposure, infrastructure leakage, dan modern attack surface.*

- SSRF fundamentals dan backend request workflow
- User-controlled input serta internal service access
- Backend request observation dengan Burp Suite
- Cloud exposure dan metadata service awareness
- Insecure cloud configuration serta sensitive data exposure
- Internal service exposure dan backend communication
- Attack surface modern pada cloud, API, dan backend

## Assessment, Defense & Karier

### 15 End-to-End Security Assessment Simulation

*Peserta mempraktikkan security assessment end-to-end: dari scope planning, testing execution, vulnerability validation, evidence collection, hingga documentation workflow.*

- Assessment planning dan scope understanding
- Target, rules of engagement, dan testing priority
- Security testing execution dari reconnaissance sampai validation
- Vulnerability findings berdasarkan assessment workflow
- Validation serta impact observation
- Evidence yang relevan untuk reporting
- Assessment documentation yang terstruktur

### 16 Professional Pentest Reporting, CVSS & Security Communication

*Peserta mempelajari penyusunan pentest report profesional, executive summary, technical findings, CVSS scoring, business impact, dan rekomendasi remediation.*

- Executive summary dan stakeholder communication
- Struktur findings documentation dalam pentest report
- Evidence, impact, dan reproduction step
- Penulisan technical findings yang terstruktur
- CVSS, severity rating, dan risk prioritization
- Business impact dari vulnerability findings
- Remediation recommendation yang actionable

### 17 SOC Fundamentals, Security Monitoring & Detection Workflow

*Peserta mempelajari dasar SOC, security monitoring, log source, detection visibility, alert triage, severity awareness, dan monitoring analysis.*

- SOC workflow dalam cybersecurity operation
- Security monitoring dan threat detection workflow
- Log source dan detection visibility
- Observasi activity yang tercatat dalam log
- Alert concept, severity, dan prioritization
- False positive dan true positive
- Detection review serta monitoring limitation analysis

### 18 Incident Analysis, IOC Investigation & Remediation Workflow

*Peserta mempelajari IOC, threat indicator, log analysis, incident investigation, root cause analysis, incident timeline reconstruction, dan rekomendasi remediation.*

- IOC fundamentals dan threat indicator analysis
- Identifikasi suspicious activity dari skenario simulasi
- Linux dan web server log analysis
- Authentication event serta suspicious log
- Incident investigation dan root cause analysis
- Incident timeline berdasarkan evidence
- Remediation dan monitoring improvement recommendation

## 19 AI-Assistant for Better Cyber Security

*Peserta mempelajari penggunaan AI untuk cybersecurity workflow, vulnerability hunting, reconnaissance analysis, technical reporting, bug bounty assessment, dan responsible disclosure.*

- Peran AI dalam cybersecurity workflow modern
- Batasan AI dan pentingnya manual validation
- AI untuk vulnerability analysis
- Exploitation flow, security impact, dan root cause
- Prompt engineering untuk cybersecurity reporting
- Bug bounty target evaluation dan assessment planning
- Vulnerability disclosure dan finding communication profesional

## 20 Job & Career: Freelance Bug Bounty & Portfolio Hacks

*Peserta mempelajari portfolio building, GitHub project showcase, pentest proposal, client communication, professional positioning, dan roadmap karier security practitioner.*

- Struktur portfolio untuk penetration testing
- Vulnerability write-up, pentest report, dan project documentation
- GitHub repository serta project showcase
- Dokumentasi tanpa membocorkan informasi sensitif
- Pentest proposal dan scope of work
- Komunikasi klien untuk non-technical stakeholder
- Professional profile serta cybersecurity roadmap

### Bootcamp Project: Website Security Assessment (2 minggu)

Peserta menjalankan assessment end-to-end dalam konteks project perusahaan, dengan weekly mentoring.

Kegiatannya mencakup scope, reconnaissance, vulnerability testing, validasi temuan, analisis risiko, dan penyusunan professional security report.

*Sumber: <https://reworkacademy.com/cyber-security-bootcamp/> | Diakses 31 Agustus 2026. Pengujian keamanan hanya boleh dilakukan pada lab atau sistem yang memiliki izin.*